

Azimuthal-Sector Protection and Exceptional-Point Search in Five-Dimensional Myers–Perry Quasinormal Spectra*

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Abstract

We study whether the enhanced symmetry of equal-magnitude-spin five-dimensional Myers–Perry black holes can organize defective degeneracies in the massive-scalar quasinormal spectrum. Away from the mass-threshold branch points, we assume a local holomorphic Fredholm realization of the quasinormal pencil that commutes with the surviving azimuthal T^2 action. The associated Fourier projectors reduce the pencil into invariant (m_1, m_2) sectors and project Jordan chains sector by sector. We prove that a coincidence of frequencies from distinct azimuthal sectors cannot increase the maximal Jordan-chain length or the order of a resolvent pole; any defectiveness must already occur within one fixed sector. The discrete symmetry is the order-eight signed-permutation group generated by independent plane reflections and plane interchange, with $a = b$ and $a = -b$ forming isometric components of the equal-magnitude-spin locus. In diagonal sectors, analytic characteristic invariants are even in $\delta = (a - b)/2$, so a quotient curve transverse to $t = \delta^2 = 0$ lifts with quadratic tangency in the remaining coordinates. We then search the surviving within-sector collision channel in a finite atlas of 50,558 continued mode entries across seven azimuthal sectors. No sampled or locally refined candidate simultaneously exhibits gap closure, one-loop branch exchange, and square-root splitting. The smallest stored branch gap is 0.5116. The numerical result is therefore a finite-atlas non-detection, not a continuum exclusion theorem.

1 Introduction

Quasinormal modes (QNMs) are resonances of black-hole perturbation operators and poles of the retarded Green function (Kokkotas & Schmidt, 1999; Berti et al., 2009; Konoplya & Zhidenko, 2011; Berti et al., 2026). Their spectral problem is non-self-adjoint because radiation leaves through the horizon and infinity. Consequently, eigenvalues may coalesce together with their eigenvectors at exceptional points (EPs) (Kato, 1995; Bergholtz et al., 2021; Ashida et al., 2020).

Let a perturbation problem be represented locally by an operator pencil $T(\omega)$. If an isolated eigenvalue ω_* has maximal Jordan-chain length q , the Keldysh expansion gives a pole of order q in $T(\omega)^{-1}$. Its inverse transform contributes

$$\psi(t) \sim (A_0 + A_1 t + \cdots + A_{q-1} t^{q-1}) e^{-i\omega_* t}. \quad (1)$$

At an EP2 the generalized mode therefore contains a $te^{-i\omega_* t}$ term. Ordinary eigenvectors alone are incomplete at the EP, while a Jordan chain supplies the appropriate generalized spectral data. Near a generic EP2, eigenvalue splitting is proportional to the square root of a transversal perturbation (Kato, 1995; Macedo et al., 2026).

Exceptional and near-exceptional structures have been identified in several black-hole spectra, including massive-scalar Kerr and model ringdown systems (Cavalcante et al., 2024; 2026; Motohashi, 2025; Macedo et al., 2026; Cao et al., 2025). Recent black-hole resonance analyses also emphasize the local Riemann-surface

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description of modes near exceptional points (Yang et al., 2025). The five-dimensional Myers–Perry geometry is attractive for three reasons. The massive Klein–Gordon equation separates for two arbitrary spins (Frolov & Stojkovic, 2003; Murata & Soda, 2008; Huang & Huang, 2025); the family has two rotational parameters plus the scalar mass; and the continuous isometry is enhanced to $\mathbb{R} \times U(2)$ when the two angular momenta have equal magnitude (Myers & Perry, 1986; Myers, 2012).

Our analysis has two complementary parts. First, we isolate the exact symmetry mechanism that constrains defectiveness. Second, we search numerically for the within-sector collisions that the theorem leaves open. The numerical claim is intentionally restricted to the computed atlas: without a continuum enclosure, a certified interpolation bound, or a rigorous radial-truncation estimate, a sampled minimum cannot be promoted to a global lower bound over a continuous parameter box.

The main results are as follows.

1. Under a strongly continuous torus action commuting with a locally holomorphic Fredholm pencil, cross-sector frequency coincidences cannot create longer Jordan chains.
2. The exact discrete symmetry is the order-eight group generated by independent plane reflections and plane interchange. The loci $a = b$ and $a = -b$ are isometric.
3. In sectors fixed by plane interchange, analytic characteristic invariants are even in $\delta = (a - b)/2$. A transversal EP curve in $t = \delta^2$ lifts to a smooth parity-symmetric physical curve whose non- δ coordinates vary quadratically through $\delta = 0$.
4. In the numerical atlas no candidate displays coalescence, branch exchange, and square-root splitting simultaneously. This is a finite-atlas non-detection result, not a continuum theorem.

2 Geometry and the separated scalar problem

2.1 Metric, horizons, and thermodynamics

In Boyer–Lindquist-type coordinates $(t, r, \theta, \phi, \psi)$, with $0 \leq \theta \leq \pi/2$ and ϕ, ψ 2π -periodic, the five-dimensional Myers–Perry metric is

$$ds^2 = -dt^2 + \frac{M}{\Sigma} (dt - a \sin^2 \theta d\phi - b \cos^2 \theta d\psi)^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + (r^2 + a^2) \sin^2 \theta d\phi^2 + (r^2 + b^2) \cos^2 \theta d\psi^2, \quad (2)$$

where

$$\Sigma = r^2 + a^2 \cos^2 \theta + b^2 \sin^2 \theta, \quad \Pi = (r^2 + a^2)(r^2 + b^2), \quad \Delta = \frac{\Pi - Mr^2}{r^2}. \quad (3)$$

The mass parameter M has dimensions of length squared. We set $M = 1$ in numerical expressions. The ADM quantities are

$$M_{\text{ADM}} = \frac{3\pi M}{8}, \quad J_\phi = \frac{2}{3} M_{\text{ADM}} a, \quad J_\psi = \frac{2}{3} M_{\text{ADM}} b. \quad (4)$$

With $z = r^2$, the horizon polynomial is

$$P(z) = (z + a^2)(z + b^2) - Mz = (z - z_+)(z - z_-), \quad (5)$$

with

$$z_\pm = \frac{1}{2} \left[M - a^2 - b^2 \pm \sqrt{(M - a^2 - b^2)^2 - 4a^2 b^2} \right]. \quad (6)$$

A horizon exists when $M \geq (|a| + |b|)^2$. We use

$$\mathcal{E} = M - (|a| + |b|)^2 \quad (7)$$

as an extremality measure. It is also useful to introduce

$$s = \frac{a+b}{2}, \quad \delta = \frac{a-b}{2}. \quad (8)$$

The equal-magnitude-spin locus $a^2 = b^2$ has two coordinate components, $\delta = 0$ and $s = 0$. A reflection of either rotation plane maps one component to the other, so they are isometric descriptions of the same equal-magnitude condition.

The horizon angular velocities, surface gravity, temperature, area, and entropy are

$$\Omega_a = \frac{a}{z_+ + a^2}, \quad \Omega_b = \frac{b}{z_+ + b^2}, \quad (9)$$

$$\kappa = \frac{r_+(z_+ - z_-)}{(z_+ + a^2)(z_+ + b^2)}, \quad T_H = \frac{\kappa}{2\pi}, \quad (10)$$

$$A_H = \frac{2\pi^2}{r_+}(z_+ + a^2)(z_+ + b^2), \quad S_{\text{BH}} = \frac{A_H}{4}. \quad (11)$$

Representative horizon data used to check the parameterization are collected in Table 1.

Table 1: Representative horizon data in units $M = 1$. The final two rows are related by interchange of the rotation planes, which swaps Ω_a and Ω_b .

s	δ	r_+	T_H	Ω_a	Ω_b
0.00	0.00	1.0000	0.15915	0.0000	0.0000
0.20	0.00	0.9583	0.15222	0.2087	0.2087
0.42	0.00	0.7713	0.11196	0.5445	0.5445
0.34	0.20	0.8249	0.12966	0.5556	0.2000
0.42	0.20	0.7296	0.10849	0.6764	0.3789
0.42	-0.20	0.7296	0.10849	0.3789	0.6764

2.2 Separation

For a scalar of mass μ , $(\square - \mu^2)\Phi = 0$, take

$$\Phi = e^{-i\omega t + im_1\phi + im_2\psi} R(r)S(\theta), \quad m_1, m_2 \in \mathbb{Z}. \quad (12)$$

The separated equations are

$$0 = \frac{1}{\sin\theta \cos\theta} \frac{d}{d\theta} \left(\sin\theta \cos\theta \frac{dS}{d\theta} \right) + \left[\Lambda + (\omega^2 - \mu^2)(a^2 \cos^2\theta + b^2 \sin^2\theta) - \frac{m_1^2}{\sin^2\theta} - \frac{m_2^2}{\cos^2\theta} \right] S, \quad (13)$$

$$0 = \frac{1}{r} \frac{d}{dr} \left(r \Delta \frac{dR}{dr} \right) + \left[\frac{W^2}{r^4 \Delta} - \frac{G^2}{r^2} - (a^2 + b^2)\omega^2 + 2\omega(am_1 + bm_2) - \mu^2 r^2 - \Lambda \right] R, \quad (14)$$

where

$$W = \Pi\omega - m_1 a(r^2 + b^2) - m_2 b(r^2 + a^2), \quad G = ab\omega - am_2 - bm_1. \quad (15)$$

At $a = b = 0$, $\Lambda = \ell(\ell + 2)$. These equations agree algebraically with the arbitrary-two-spin formulation in Huang & Huang (2025), after accounting for the convention used for the separation constant.

Set

$$\hat{A} = \Lambda + (\omega^2 - \mu^2)b^2, \quad c^2 = (\omega^2 - \mu^2)(a^2 - b^2) = 4s\delta(\omega^2 - \mu^2). \quad (16)$$

When $a^2 = b^2$, $c^2 = 0$ and the angular operator is the round scalar Laplacian on S^3 ,

$$\hat{A} = \ell(\ell + 2), \quad \ell = 2n + |m_1| + |m_2|, \quad n \in \mathbb{Z}_{\geq 0}. \quad (17)$$

The angular eigenspace has dimension $(\ell + 1)^2$. For the full rotating spacetime, the relevant continuous isometry on either equal-magnitude component is $U(2)$ rather than the full static $SO(4)$ symmetry (Myers, 2012).

At the outer horizon,

$$\sigma_+ = \frac{\omega - m_1\Omega_a - m_2\Omega_b}{2\kappa} = \frac{\omega - \omega_c}{2\kappa}, \quad \omega_c = m_1\Omega_a + m_2\Omega_b, \quad (18)$$

and the QNM boundary condition selects

$$R \sim (z - z_+)^{-i\sigma_+}. \quad (19)$$

At large r ,

$$R'' + \frac{3}{r}R' + (\Omega^2 + O(r^{-2}))R = 0, \quad \Omega = \sqrt{\omega^2 - \mu^2}, \quad (20)$$

so the outgoing branch is

$$R \sim r^{-3/2}e^{+i\Omega r}(1 + O(r^{-1})). \quad (21)$$

There is no $1/r$ Coulomb tail in this five-dimensional radial equation.

2.3 Superradiance and the local analytic pencil

For a real-frequency scattering state, propagation at infinity requires $\Omega \in \mathbb{R}$, hence $\omega > \mu$ for positive frequency. The horizon flux has the sign of $\omega - \omega_c$. Scattering amplification therefore occurs for

$$\mu < \omega < \omega_c. \quad (22)$$

For a quasibound state, the field instead decays at infinity and a superradiant instability may occur when $0 < \text{Re } \omega < \omega_c$ together with $\text{Re } \omega < \mu$ (Detweiler, 1980; Dolan, 2007). The ingoing horizon condition (19) remains the physical horizon condition on both sides of $\omega = \omega_c$; what changes is the flux sign, not the definition of ingoing propagation.

The boundary condition at infinity contains the square root $\Omega = \sqrt{\omega^2 - \mu^2}$. Accordingly, a holomorphic pencil description is local: one chooses a sheet of the square root and works away from the threshold branch points $\omega = \pm\mu$. On such a neighborhood the coupled angular-radial problem may be written

$$T(\omega; s, \delta, \mu, m_1, m_2)u = 0, \quad (23)$$

with $\Lambda(\omega)$ recomputed at every evaluation. Derivatives with respect to ω must be total derivatives; freezing Λ drops the term $(\partial_\Lambda F)(d\Lambda/d\omega)$ and can create spurious multiple-root diagnostics.

3 Azimuthal-sector protection theorem

3.1 Torus action and projectors

Neither the metric nor the boundary conditions depend explicitly on ϕ or ψ . Define the torus action

$$(R_{\alpha\beta}u)(t, r, \theta, \phi, \psi) = u(t, r, \theta, \phi - \alpha, \psi - \beta). \quad (24)$$

A mode proportional to $e^{im_1\phi + im_2\psi}$ transforms with character $e^{-i(m_1\alpha + m_2\beta)}$. Therefore the projector onto the sector labeled by the ansatz (12) is

$$\Pi_{m_1 m_2} = \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} e^{+i(m_1\alpha + m_2\beta)} R_{\alpha\beta} d\alpha d\beta. \quad (25)$$

The plus sign in Eq. (25) is essential for the convention (24); the opposite sign projects onto $(-m_1, -m_2)$.

Proposition 1 (Invariant sector decomposition). *Let X be a Banach space on which $R_{\alpha\beta}$ is a uniformly bounded, strongly continuous representation of T^2 . Suppose that $T(\omega)$ and its local ω derivatives commute with the action and that the operator domain is invariant. Then the projectors (25) are bounded, pairwise annihilating idempotents that commute with the pencil. Moreover,*

$$\Pi_{m_1 m_2} u = 0 \text{ for all } (m_1, m_2) \implies u = 0. \quad (26)$$

Proof. The Bochner integral exists by uniform boundedness. Idempotence and pairwise annihilation follow from character orthogonality. To show separation of points without assuming unconditional Fourier-series convergence, convolve the orbit map with the two-dimensional Fejér kernel. The Fejér sums are finite positive combinations of the projectors and converge in norm to u by strong continuity. If all projected components vanish, every Fejér sum vanishes and hence $u = 0$. $\square$

The statement is local in spectral parameter. We assume a weighted or hyperboloidal realization for which the quasinormal pencil is holomorphic and Fredholm on a chosen sheet away from $\omega = \pm\mu$, and for which the torus action preserves the operator domain. For an isolated characteristic value, the root subspace is finite dimensional; its torus representation therefore contains only finitely many characters (Güttel & Tisseur, 2017). No global Fredholm construction at the mass threshold is asserted.

3.2 Jordan chains and pole order

A Jordan chain $(v_0, \dots, v_{q-1})$, with $v_0 \neq 0$, satisfies

$$\sum_{i=0}^j \frac{1}{i!} T^{(i)}(\omega_*) v_{j-i} = 0, \quad j = 0, \dots, q-1. \quad (27)$$

Lemma 1. *Under the preceding hypotheses, every Jordan chain projects componentwise to Jordan chains of the sector restrictions. At least one projected chain has the same length, and a chain in one sector extends by zero to the full space. Thus the maximal chain length of T is the maximum of the sectorwise maximal chain lengths.*

Proof. Apply $\Pi_{m_1 m_2}$ to every equation in (27). Commutation with $T^{(i)}$ gives the sector chain equations. Since $v_0 \neq 0$, the separation property implies that at least one $\Pi_{m_1 m_2} v_0$ is nonzero. The converse extension is immediate. $\square$

Theorem 1 (Cross-sector no-go). *Assume the local holomorphic Fredholm realization described above, and let ω_* be an isolated characteristic value. The maximal Jordan-chain length at ω_* equals the maximum of the maximal chain lengths of the sector restrictions. Hence a frequency coincidence between distinct azimuthal sectors cannot, by itself, create defectiveness, raise the resolvent-pole order, or generate an additional secular power of t in Eq. (1). Any chain of length greater than one is contained in a single fixed (m_1, m_2) sector.*

The wording does not exclude a frequency shared by two sectors when one sector independently contains an EP. In that case the defect belongs to the single sector and is not caused by the cross-sector coincidence.

3.3 Equal-magnitude-spin multiplets

On $a = b$ the radial equation depends on the azimuthal labels through $m = m_1 + m_2$. At fixed (ℓ, m) , the degenerate labels are

$$\mathcal{M}_{\ell, m} = \{(m_1, m_2) : m_1 + m_2 = m, |m_1| + |m_2| \leq \ell, \ell - |m_1| - |m_2| \in 2\mathbb{Z}_{\geq 0}\}. \quad (28)$$

For each admissible $m = -\ell, -\ell + 2, \dots, \ell$, one finds $|\mathcal{M}_{\ell, m}| = \ell + 1$. Summing over the $\ell + 1$ values of m gives the angular degeneracy $(\ell + 1)^2$. Members of a fixed $\mathcal{M}_{\ell, m}$ occupy distinct azimuthal sectors, so their coincidence is semisimple across sectors unless a participating sector is independently defective.

Table 2: Fixed-sector stabilizers of the signed-permutation symmetry. Spectral equalities are setwise; an isolated simple branch can be labeled pointwise.

Sector	Stabilizer	Parameter action	Spectral relation
$m_1 = m_2$	E	$(s, \delta) \mapsto (s, -\delta)$	$\text{spec}_{m,m}(s, \delta) = \text{spec}_{m,m}(s, -\delta)$
$m_1 = -m_2$	EAB	$(s, \delta) \mapsto (-s, \delta)$	$\text{spec}_{m,-m}(s, \delta) = \text{spec}_{m,-m}(-s, \delta)$
$m_1 = 0$	A	$(s, \delta) \mapsto (-\delta, -s)$	$\text{spec}_{0,m_2}(s, \delta) = \text{spec}_{0,m_2}(-\delta, -s)$
$m_2 = 0$	B	$(s, \delta) \mapsto (\delta, s)$	$\text{spec}_{m_1,0}(s, \delta) = \text{spec}_{m_1,0}(\delta, s)$
$(0, 0)$	G	all signed permutations	full induced invariance

The continuation confirms this directly. On the slice $\ell = 2$, $N = 0$, $s = 0.20$, $\mu = 0.6$, the $(1, 1)$ and $(2, 0)$ branches agree at $\delta = 0$ to $\text{Re}\omega = 1.6835979353$ in both sectors, coinciding to 4×10^{-15} , and separate to 1.6892638 and 1.8006703 at $\delta = 0.20$. The diagonal sector is even in δ to machine precision, returning 1.6892638 at $\delta = -0.20$ as well, while the off-diagonal sector maps to its partner $(0, 2)$ and returns 1.5994653. No spectral surface is inferred outside this computed slice. The theorem protects against cross-sector hybridization, not against spectral splitting.

The component $a = -b$ is obtained from $a = b$ by reflecting one azimuthal plane. Its radial multiplets are equivalently organized by $m_1 - m_2$ after the corresponding label reflection.

4 Discrete symmetries and local exceptional topology

4.1 The full signed-permutation group

In addition to interchange of the two rotation planes, the scalar problem is invariant under independent orientation reversals:

$$A : (a, m_1, \phi) \mapsto (-a, -m_1, -\phi), \quad (29)$$

$$B : (b, m_2, \psi) \mapsto (-b, -m_2, -\psi), \quad (30)$$

$$E : (a, b, m_1, m_2, \theta, \phi, \psi) \mapsto (b, a, m_2, m_1, \frac{\pi}{2} - \theta, \psi, \phi). \quad (31)$$

They generate

$$G = \langle A, B, E \rangle \simeq (\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes S_2, \quad |G| = 8. \quad (32)$$

The subgroup $\{1, E, AB, EAB\}$ is a Klein four-group, but it is not the full discrete symmetry. In (s, δ) coordinates,

$$E : (s, \delta) \mapsto (s, -\delta), \quad AB : (s, \delta) \mapsto (-s, -\delta), \quad (33)$$

$$A : (s, \delta) \mapsto (-\delta, -s), \quad B : (s, \delta) \mapsto (\delta, s). \quad (34)$$

Table 2 lists the nontrivial fixed-sector symmetries. The equalities are naturally setwise statements for the spectrum; an isolated simple branch can be labeled to satisfy them pointwise.

4.2 Discriminant, semisimple coincidences, and the $t = \delta^2$ quotient

A local two-mode spectral subspace may be represented by a 2×2 matrix $H(p)$. The similarity-invariant quantities are

$$\mathcal{T} = \text{tr } H, \quad \mathcal{D} = (\text{tr } H)^2 - 4 \det H, \quad (35)$$

with eigenvalues

$$\omega_{\pm} = \frac{1}{2}(\mathcal{T} \pm \sqrt{\mathcal{D}}). \quad (36)$$

The equation $\mathcal{D} = 0$ gives a repeated algebraic eigenvalue. It is not, by itself, equivalent to an EP: the special case $H = \omega_* I$ is a semisimple double eigenvalue. A simple zero of $\mathcal{D}$ in a transversal direction is generically

defective and yields the square-root Puiseux behavior

$$\omega_{\pm}(\tau) = \omega_* + c_1\tau \pm c_{1/2}\tau^{1/2} + O(\tau^{3/2}), \quad c_{1/2} \neq 0. \quad (37)$$

Geometric multiplicity or a Jordan-chain test must still be checked at the repeated root.

In a diagonal sector, the characteristic polynomial and its analytic invariants are even in δ . Individual eigenvalue labels need not be analytic or separately even at an EP. Let $t = \delta^2$ and suppose an EP component in quotient coordinates (s, t, μ) meets $t = 0$ transversely. With a local curve parameter τ ,

$$t = c\tau + O(\tau^2), \quad c \neq 0. \quad (38)$$

For any other smooth coordinate q along the curve (for example s or μ), write $q = q_* + A\tau + O(\tau^2)$. Eliminating τ gives

$$\tau = \frac{\delta^2}{c} + O(\delta^4), \quad q - q_* = \frac{A}{c}\delta^2 + O(\delta^4). \quad (39)$$

The two quotient preimages therefore join into a single smooth parity-symmetric curve in the full physical parameter space. At $\delta = 0$ the tangent direction is along the δ coordinate, while the remaining coordinates have a parabolic profile. The quotient locus has an endpoint at $t = 0$; the physical curve does not terminate there. Equation (39) is the local normal form of this lift.

5 Exceptional-point diagnostics

A credible negative search requires diagnostics that are invariant under harmless reformulations of the spectral condition and that are calibrated on positive and negative controls.

5.1 Admissible signatures

For a generic EP2, the following signatures are complementary:

1. independently tracked eigenvalue separation tends to zero under local refinement;
2. one circuit around the candidate transposes the two branches, and two circuits restore them;
3. the local splitting is better fit by $|\tau|^{1/2}$ than by $|\tau|$ or a nonzero constant (Nennig & Perrey-Debain, 2020);
4. the repeated root has geometric multiplicity one, or an associated vector solves the Jordan-chain equation;
5. the result is stable under discretization, precision, and a numerically independent formulation.

For a meromorphic scalar condition $F(\omega)$, the argument principle gives

$$\frac{1}{2\pi i} \oint_C \frac{F'(\omega)}{F(\omega)} d\omega = N_Z - N_P. \quad (40)$$

Zeros and poles must therefore be separated, rather than interpreting $N_Z - N_P$ as a zero count when poles are present.

A matrix-based nonlinear-eigenvalue condition indicator may be written

$$\kappa_T(\omega_*) = \frac{\|u\| \|v\|}{|\langle v, T'(\omega_*)u \rangle|}, \quad (41)$$

where u and v are right and left null vectors and v belongs to the dual space in the operator setting. In a fixed matrix normalization, growth of κ_T is a useful qualitative warning. It is not an absolute distance-to-EP bound: under $T \mapsto g(\omega)T$ with $g(\omega_*) \neq 0$, Eq. (41) scales by $|g(\omega_*)|^{-1}$ unless a normalization convention is fixed.

Table 3: Diagnostic calibration on exactly solvable controls. The fitted exponent separates square-root, linear, and floored splitting; the normalization test shows that the coefficient ratio is not an invariant proximity measure, whereas the argument-principle zero count is.

<i>Local splitting laws:</i> fitted Puiseux exponent p over $t \in [10^{-8}, 10^{-4}]$			
Model		p	Classification
square root, $\omega_{\pm} = \pm\sqrt{t}$		0.500	EP2
linear, $\omega_{\pm} = \pm 3t$		1.000	ordinary crossing
floored, $2\sqrt{t^2 + g^2}$		0.000	avoided crossing
<i>Normalization sensitivity</i> under $F \mapsto gF$, $g = 10^7 e^{3\omega}$			
d	measured distortion	$ 1 + (g'/g)d ^{-1}$	zero count
1×10^{-4}	1.000	1.000	2
1×10^{-2}	1.031	1.031	2
3×10^{-1}	10.000	10.000	2

5.2 Normalization-sensitive diagnostics

The derivative $|F'(\omega_*)|$ of an arbitrarily normalized scalar spectral condition has no invariant magnitude. Likewise, if

$$F = a_1(\omega - \omega_1) + a_2(\omega - \omega_1)^2 + \dots, \quad (42)$$

then under $F \mapsto gF$, with $g = g_0 + g_1(\omega - \omega_1) + \dots$,

$$\frac{\tilde{a}_1}{\tilde{a}_2} = \frac{a_1/a_2}{1 + (g_1/g_0)(a_1/a_2)}. \quad (43)$$

The coefficient ratio is only asymptotically insensitive to normalization. More importantly, for a meromorphic continued fraction it responds to the nearest zero or pole. It cannot serve as a rigorous spectral separation bound.

Table 3 records the calibration of these diagnostics on exactly solvable controls, computed with the same routines that are applied to the atlas. The fitted Puiseux exponent cleanly separates the three local laws, and the coefficient-ratio diagnostic inflates by an order of magnitude under a rescaling that leaves the zero count exactly invariant. Calibration and candidate evidence play different roles in an EP classification and are kept apart throughout.

6 Numerical formulations and validation

The numerical calculation combines recurrence-based and recurrence-free formulations. Implementation details, atlas census information, and diagnostic checks are summarized in the Supplementary Material.

6.1 Continued fraction

Introduce the compact radial coordinate $x = (z - z_+)/ (z - z_-)$ and factor the ingoing outer-horizon behavior and outgoing large- r behavior from the radial solution. The remaining analytic factor is expanded in powers of x . The radial Frobenius expansion yields a finite-band recurrence

$$\sum_{j=0}^p c_j(n) a_{n-j} = 0, \quad p + 1 = 13, \quad (44)$$

for generic two-spin parameters, reducing to width 9 in the static case. Gaussian elimination gives a three-term recurrence and a Leaver continued fraction (Leaver, 1985). The recurrence coefficients are generated

Table 4: Numerical formulations and their role in the validation strategy. Solvers A and B share the Frobenius recurrence; C and D are recurrence-free collocation formulations.

Solver	Formulation	Recurrence-free	Primary role / limitation
A	Gaussian-reduced Leaver continued fraction	no	Main continuation solver; finite-depth drift near threshold
B	Raw recurrence, Hill determinant, Wynn acceleration	no	Same recurrence family as A; not an independent cross-check
C	Exterior-complex-scaled Chebyshev collocation	yes	Independent resonance check when an admissible scaled contour exists
D	Multidomain near-horizon collocation	yes	Cross-check for ordinary modes; shares the outgoing-contour threshold obstruction

algorithmically by Fourier coefficient extraction as described in the Supplementary Material. Because the coefficient extraction and radial truncation are not accompanied by a rigorous aliasing/truncation enclosure, the continued-fraction calculation is used as numerical evidence rather than as a certified spectral enclosure. The inversion index must follow the tracked overtone: finite-depth inversions are not numerically interchangeable. The large- n ratio used for tail acceleration is

$$\frac{a_n}{a_{n-1}} = 1 + u_1 n^{-1/2} + u_2 n^{-1} + O(n^{-3/2}), \quad u_1 = -\sqrt{-2c}, \quad c = i\Omega(r_+ - r_-), \quad (45)$$

with the square-root sign selected by minimality.

6.2 Exterior complex scaling

A recurrence-free formulation rotates the radial contour beyond r_m ,

$$r = r_m + \varsigma e^{i\vartheta}, \quad 0 \leq \varsigma \leq L. \quad (46)$$

The outgoing factor decays along the contour when

$$\tan \vartheta > -\frac{\text{Im } \Omega}{\text{Re } \Omega}, \quad (47)$$

for $\text{Re } \Omega > 0$. Resonances should be stable under changes of ϑ , whereas the rotated continuum moves.

6.3 Solver inventory and benchmarks

The solver inventory is summarized in Table 4. Solvers A and B share the same recurrence and are not independent. Solver C is the principal recurrence-free cross-check. The validation suite reproduces the five-dimensional Schwarzschild–Tangherlini scalar fundamental mode

$$\omega = 0.533835574268 - 0.383375368513 i \quad (r_+ = 1) \quad (48)$$

within 5.0×10^{-13} of Matyjasek (2021), exchange symmetry to 4.2×10^{-15} , and three-solver agreement below 10^{-5} on ten benchmark points; comparable spectral-method cross-checks are surveyed by Hatsuda & Kimura (2021) and Stein (2019). These checks support implementation consistency at selected points, but they do not bound the numerical error throughout the continuous parameter domain.

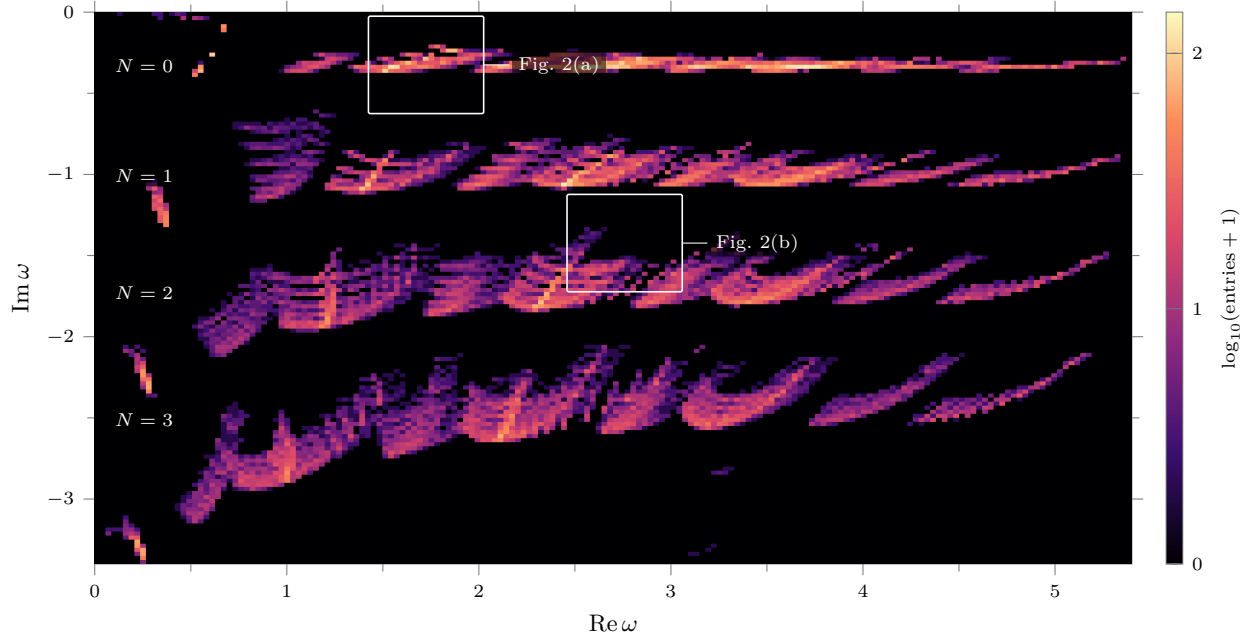


Figure 1: The stored quasinormal atlas. Occupancy of all 50,558 stored mode entries, across seven azimuthal sectors and 84 continued branches, on a 180×136 grid of the complex-frequency plane; the cell value is $\log_{10}(\text{entries} + 1)$, so an empty cell is exactly empty. The overtone ladders $N = 0, 1, 2, 3$ are labelled at their median damping. Boxes mark the two windows opened in Figure 2. Only 20% of the plotted cells contain any stored entry: the search is finite, and the figure shows exactly where it went.

7 Finite-atlas search and numerical result

7.1 Atlas construction

Branches are labeled by continuation history rather than instantaneous sorting. The continuation acceptance rule for the next predictor miss is

$$\epsilon_{k+1} \leq \max[4 \text{median}(\epsilon_1, \dots, \epsilon_k), 0.02 \max(|\omega_{k+1}|, 1)]. \quad (49)$$

The finite atlas contains 50,558 mode entries in the seven independent sectors

$$(0, 0), (1, 1), (-1, -1), (1, 0), (2, 0), (2, 1), (2, 2), \quad (50)$$

with

$$\ell \in \{\ell_{\min}, \ell_{\min} + 2, \ell_{\min} + 4\}, \quad \ell_{\min} = |m_1| + |m_2|, \quad N = 0, 1, 2, 3. \quad (51)$$

The declared sampled region is

$$0 \leq s \leq 0.42, \quad |\delta| \leq 0.20, \quad 0 \leq \mu \leq 1.8, \quad r_- \leq 0.2287, \quad \mathcal{E} \geq 0.294. \quad (52)$$

Table 5 summarizes the continuation census, including predictor misses, nonconvergences, and excluded branch-label collapses. Figure 1 shows the stored atlas itself in the complex-frequency plane. The four horizontal ladders are the overtones $N = 0, 1, 2, 3$; the filamentary texture within each ladder is the trace of the continuation sweeps in (s, δ, μ) , and the empty regions between them are genuinely empty rather than merely unresolved.

7.2 What the atlas establishes

Over the stored sample, the extrema recorded in the numerical summary are

$$a_{\min}^{(\text{sample})} = 0.5116, \quad \kappa_{\max}^{(\text{fixed scaling})} = 8.73 \times 10^3. \quad (53)$$

Table 5: Finite-atlas census and continuation bookkeeping. Counts refer to stored computations; they do not constitute a covering certificate for the continuous parameter domain.

Quantity	Value
Stored mode entries	50,558
Multi-branch parameter points	4,459
Independent azimuthal sectors	7
Angular branches per sector	3
Overtones	$N = 0, 1, 2, 3$
Maximum stored continued-fraction residual	3.6×10^{-9}
Predictor misses	118
Nonconvergences	33
Branch-label collapses excluded	4

The minimum occurs on the boundary of the sampled region. Relative to the stated control value 18.3, the condition indicator is larger by approximately

$$\frac{8.73 \times 10^3}{18.3} \simeq 4.77 \times 10^2, \quad (54)$$

not 476 exactly. This elevation is consistent with a close avoided crossing in the chosen matrix normalization, but it is not a normalization-independent distance bound.

Figure 2 shows what that elevation looks like as a geometric object. The resolvent landscape at the tightest stored interaction is visibly more distorted than at the ordinary control, but the level sets close around a single tracked mode rather than pinching towards a common point, which is the signature a genuine coalescence would leave.

The ten closest stored pairs remain separated under the documented local refinements and show neither one-loop branch exchange nor $t^{1/2}$ splitting. These observations favor avoided crossings. Candidate-by-candidate recurrence-free confirmation is not available for every member of this set, so the classification is kept provisional. The numerical conclusion is therefore: no exceptional point was detected among the sampled and locally refined candidates. This statement is restricted to the computed set and does not exclude EPs throughout the continuous domain in Eq. (52).

8 Threshold and quasisresonant behavior

A separate continuation follows eleven branches in six sectors as μ increases. Along one branch, the damping decreases from 0.357 at $\mu = 0$ to 8.65×10^{-10} at $\mu = 2.968$, while $\text{Re } \omega \rightarrow \mu$. The neighboring overtone remains at damping of order 10^{-1} , so the observed behavior is not a two-mode coalescence in the tracked window.

Exterior complex scaling becomes ineffective in this limit because Eq. (47) requires

$$\vartheta_{\min} \rightarrow 90^\circ \quad \text{as} \quad \text{Re } \Omega \rightarrow 0. \quad (55)$$

This is a limitation of the two recurrence-free formulations used here, not a proof that no recurrence-free or hyperboloidal method can treat the threshold. A hyperboloidal formulation is a promising route into this region (Jaramillo et al., 2021; Macedo et al., 2026).

No full-atlas superradiant margin can be inferred from the aggregate summary statistics alone. In particular, Table 1 gives $\Omega_a = \Omega_b = 0.5445$ at $s = 0.42$, $\delta = 0$, so the $(1, 1)$ critical frequency there is approximately 1.089, not bounded by 0.51. Superradiant statements must be evaluated branch by branch and must distinguish scattering QNMs from decaying quasibound states.

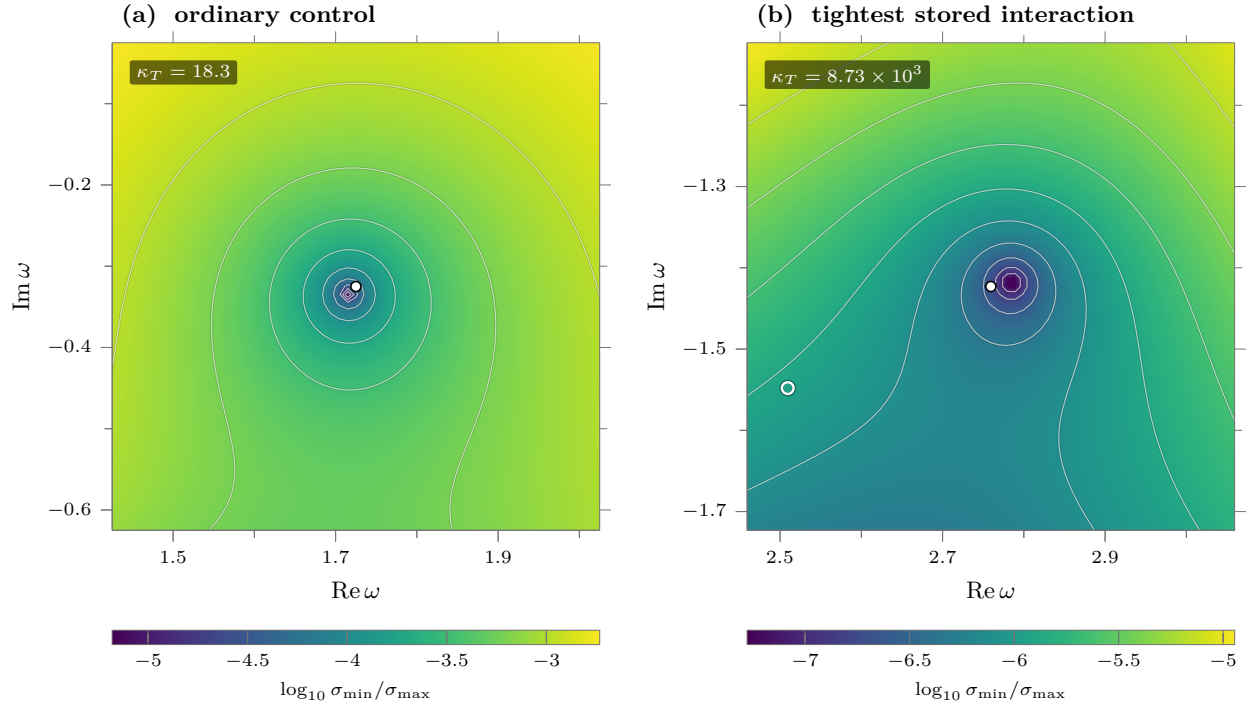


Figure 2: Pseudospectral landscape of the quasinormal pencil. $\log_{10} \sigma_{\min}/\sigma_{\max}$ of the Solver C exterior-complex-scaled collocation matrix on 61×61 complex-frequency grids, with level sets extracted from the same fields. Each panel is centred on its own tracked mode and both use the same window size and the same square aspect ratio, so the two landscapes are directly comparable. (a) An ordinary control at $s = 0.24$, $\delta = 0.10$, $\mu = 0.6$ in sector $(1, 1)$, $\ell = 2$. (b) The tightest stored interaction, at the corner $s = 0.42$, $\delta = -0.20$, $\mu = 1.8$ in sector $(2, 0)$, $\ell = 4$, whose condition indicator is the atlas maximum $\kappa_T = 8.73 \times 10^3$. Filled markers are the tracked modes. The open marker in (b) is the nearer member of the closest stored pair; its partner lies below the plotted window at $\text{Im } \omega = -2.055$, the two being separated by $d_{\min}^{(\text{sample})} = 0.5116$. Note that the two colour scales differ: the field at (b) runs about two decades deeper than at (a) over a comparable span, so the landscape there is far more sensitive. Its level sets nonetheless remain organized around one mode rather than pinching towards a shared point. Both fields are properties of the finite-dimensional discretization; no continuum pseudospectral claim is made.

9 Physical interpretation and limitations

The five-dimensional asymptotically flat Myers–Perry black hole is not an astrophysical model of observed four-dimensional black holes. Its value here is structural. The sector theorem applies whenever a valid QNM realization commutes with surviving azimuthal symmetries. In four-dimensional Kerr, for example, modes of different m lie in distinct invariant sectors; the known massive-scalar EPs occur within a fixed m sector (Cavalcante et al., 2024; 2026).

The higher-dimensional instability context must also be stated carefully. Equal-angular-momentum Myers–Perry black holes were found to show no gravitational instability in five or seven dimensions in the axisymmetric analysis of Dias et al. (2010); the instability in that axisymmetric analysis first appeared in nine dimensions. Other nonaxisymmetric and ultraspinning instabilities occur in higher-dimensional families and dimensions (Hartnett & Santos, 2013; Emparan et al., 2014), but the scalar QNM calculation here does not establish gravitational stability or instability of the five-dimensional background.

The principal limitations are:

1. the theorem is conditional on a specified local holomorphic Fredholm realization and excludes only cross-sector mechanisms;

2. the numerical search is finite and the radial recurrence has no certified truncation/aliasing enclosure;
3. overtones $N > 3$, higher angular branches, $s > 0.42$, $\mu > 1.8$ in the atlas, and the superradiant bound-state regime were not searched;
4. the threshold region is not recurrence-free verified by the formulations used here.

10 Discussion and conclusions

The central conclusion is structural. Equal-magnitude-spin symmetry produces exact frequency degeneracies, but the surviving azimuthal torus symmetry simultaneously prevents states in different (m_1, m_2) sectors from forming a shared defective block. Exceptional behavior therefore remains possible only through collisions of distinct spectral branches inside one fixed azimuthal sector.

No such exceptional collision was detected in the finite atlas. The smallest stored gap is 0.5116, and the closest candidates show neither branch exchange nor square-root splitting under the documented local tests. Because the calculation does not include a certified covering or radial-truncation enclosure, this result is a finite-atlas non-detection, not a theorem excluding exceptional points throughout the continuous search box.

Two methodological lessons are broadly useful. First, derivatives and Taylor coefficients of an arbitrarily normalized meromorphic spectral condition are not absolute proximity measures. Second, a small gap between numerical labels is evidence about branch tracking before it is evidence about physics; continuation histories, independent formulations, monodromy, and local Puiseux fits are necessary safeguards.

Data availability

The numerical continuation atlas and solver materials supporting the finite-atlas claims are available from the authors upon reasonable request through the anonymized review process. They are not publicly archived at the time of submission. The anonymized Supplementary Material contains the solver inventory, atlas census, verification gate, symmetry checks, and threshold summary.

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A Checks on separation and singular structure

The determinant of Eq. (2) satisfies

$$\sqrt{|g|} = r \Sigma \sin \theta \cos \theta. \quad (56)$$

Substitution of Eq. (12) into

$$\partial_\nu (\sqrt{|g|} g^{\nu\rho} \partial_\rho \Phi) = \mu^2 \sqrt{|g|} \Phi \quad (57)$$

separates into Eqs. (13) and (14). In $z = r^2$, the potential has apparent singularities at $z = 0, z_+, z_-$. When $ab \neq 0$, the apparent $1/z$ terms cancel because $W(0) = abG$ and $P(0) = a^2b^2$. The physical regular singular points are $z_\pm$, and infinity is an irregular singular point of rank one.

At $z = z_+$, factorization gives

$$W(z_+) = (z_+ + a^2)(z_+ + b^2)(\omega - \omega_c), \quad (58)$$

leading to Eq. (19). At infinity, division of Eq. (14) by its leading radial coefficient gives the asymptotic equation preceding Eq. (21). The absence of an $O(r^{-1})$ term removes the four-dimensional Coulomb logarithm.

B Keldysh pole order

Let $T(\omega)$ be holomorphic and Fredholm of index zero in a neighborhood of an isolated characteristic value ω_* . For a simple eigenvalue, with right and left null vectors normalized by $\langle v, T'(\omega_*)u \rangle = 1$,

$$T(\omega)^{-1} = \frac{u \otimes v}{\omega - \omega_*} + O(1). \quad (59)$$

For maximal chain length q , the local Keldysh expansion has

$$T(\omega)^{-1} = \sum_{k=1}^q \frac{S_k}{(\omega - \omega_*)^k} + O(1), \quad S_q \neq 0. \quad (60)$$

Thus the pole order equals the maximal partial multiplicity. Inverse Laplace transformation of a pole of order k produces $t^{k-1}e^{-i\omega_*t}/(k-1)!$, giving Eq. (1). Combined with the sector lemma, this proves the stated relation between cross-sector coincidences and ringdown pole order.